

HILL CIPHER

Aim

To implement the **Hill Cipher encryption technique** using a key matrix and matrix multiplication in C.

Algorithm

1. **Start** the program.
2. Read the plaintext message.
3. Convert the plaintext characters to uppercase.
4. Convert each alphabet character into a number using: A = 0, B = 1, C = 2, ..., Z = 25
5. Select a key matrix. For example:
6. Divide the plaintext into blocks of two characters.
7. Multiply each plaintext block with the key matrix.
8. Apply modulo 26 to the resulting values.
9. Convert the resulting numbers back into alphabets.
10. Display the encrypted ciphertext.
11. **Stop** the program

Program :

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

int main()
{
    char plaintext[100];
    char ciphertext[100];
    int key[2][2] = {{3, 3}, {2, 5}};
    int i, x, y;

    printf("Enter the plaintext: ");
    scanf("%s", plaintext);

    // Convert plaintext to uppercase
    for (i = 0; plaintext[i] != '\0'; i++)
```

```

{
    plaintext[i] = toupper(plaintext[i]);
}

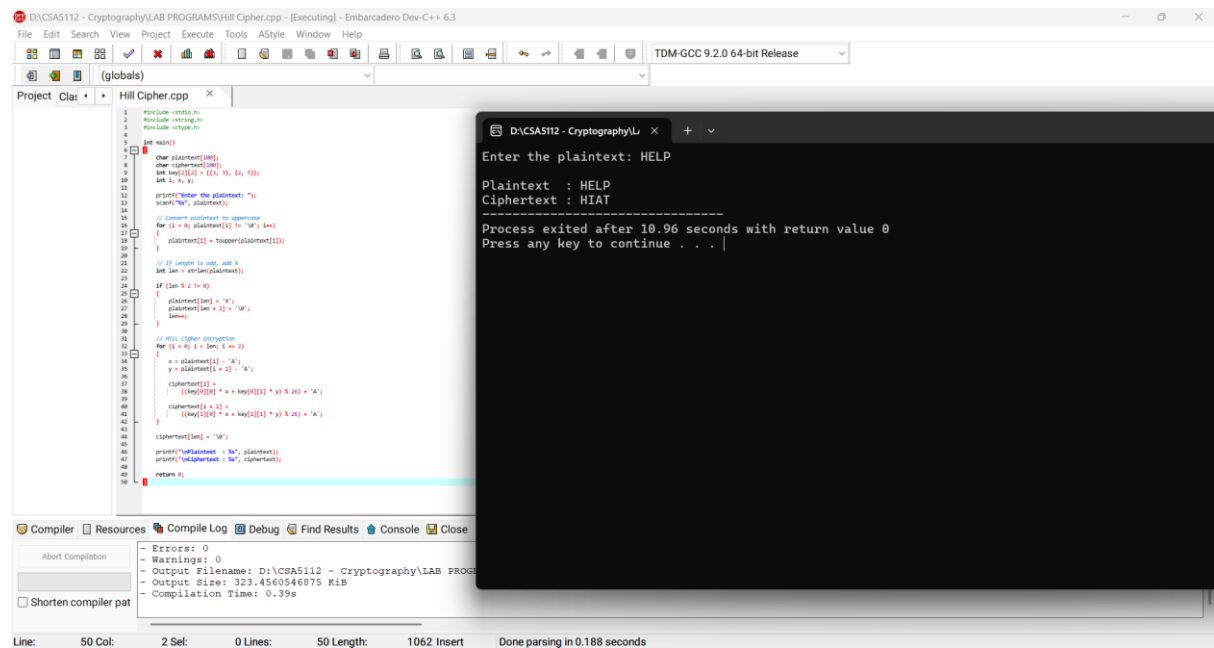
// If length is odd, add X
int len = strlen(plaintext);

if (len % 2 != 0)
{
    plaintext[len] = 'X';
    plaintext[len + 1] = '\0';
    len++;
}

// Hill Cipher Encryption
for (i = 0; i < len; i += 2)
{
    x = plaintext[i] - 'A';
    y = plaintext[i + 1] - 'A';
    ciphertext[i] =
        ((key[0][0] * x + key[0][1] * y) % 26) + 'A';
    ciphertext[i + 1] =
        ((key[1][0] * x + key[1][1] * y) % 26) + 'A';
}
ciphertext[len] = '\0';
printf("\nPlaintext : %s", plaintext);
printf("\nCiphertext : %s", ciphertext);
return 0;
}

```

OUTPUT :



```
1 #include <iostream>
2 #include <string>
3 #include <cstring>
4
5 int main()
6 {
7     char plaintext[100];
8     char ciphertext[100];
9     int key[3][3] = {{1, 0, 0}, {0, 1, 0}};
10     int i, n, j;
11
12     printf("Enter the plaintext: ");
13     scanf("%s", plaintext);
14
15     // Convert plaintext to uppercase
16     for (i = 0; i < strlen(plaintext); i++)
17     {
18         plaintext[i] = toupper(plaintext[i]);
19     }
20
21     // If length is odd, add 'X'
22     int len = strlen(plaintext);
23     if (len % 2 != 0)
24     {
25         plaintext[len] = 'X';
26         plaintext[len + 1] = '\0';
27         len++;
28     }
29
30     // Hill cipher encryption
31     for (i = 0; i < len; i += 2)
32     {
33         x = plaintext[i] - 'A';
34         y = plaintext[i + 1] - 'A';
35
36         ciphertext[i] = (x * key[0][0] + y * key[0][1]) % 26;
37         ciphertext[i + 1] = (x * key[1][0] + y * key[1][1]) % 26;
38     }
39
40     ciphertext[len] = '\0';
41     printf("Plaintext : %s", plaintext);
42     printf("Ciphertext : %s", ciphertext);
43
44     return 0;
45 }
```

```
D:\CSA5112 - Cryptography\LAB PROGRAMS\Hill Cipher.cpp
Enter the plaintext: HELP
Plaintext : HELP
Ciphertext : HIAT
-----
Process exited after 10.96 seconds with return value 0
Press any key to continue . . .
```

Compiler | Resources | Compile Log | Debug | Find Results | Console | Close

Abort Compilation

Shorten compiler path

Errors: 0
Warnings: 0
Output Filename: D:\CSA5112 - Cryptography\LAB PROGRAMS\Hill Cipher.cpp
Output Size: 323.4560546875 KiB
Compilation Time: 0.39s

Line: 50 Col: 2 Sel: 0 Lines: 50 Length: 1062 Insert Done parsing in 0.188 seconds